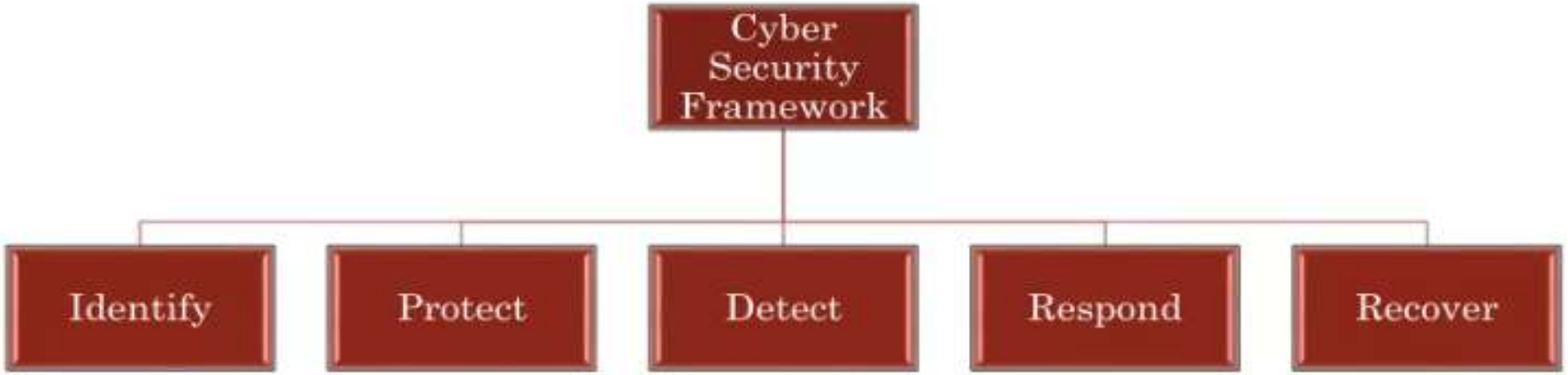
SOC TRAINING

OUTLINE

- Cyber Security Framework
- What is SOC ?
- SOC Team
- SOC process
- SOC Platform (Tools)
- Skills needed in a SOC
- Types of SOCs

CYBER SECURITY FRAMEWORK



CYBER SECURITY FRAMEWORK *(CONT.)*

- Identify
  - Identify threats which needed to protect our enterprise.
  - Control who can access your business information.
  - Require individual user accounts for each employee.
  - Create policies and procedures.
- Protect
  - Install and activate security controls (Firewalls, IDS/IPS, ….).
  - Patch your operating systems and applications routinely.
  - Secure your wireless access point and networks.
  - Setup web and E-mail filters.
  - Use encryption for sensitive data.
  - Train employees for security awareness.

CYBER SECURITY FRAMEWORK *(CONT.)*

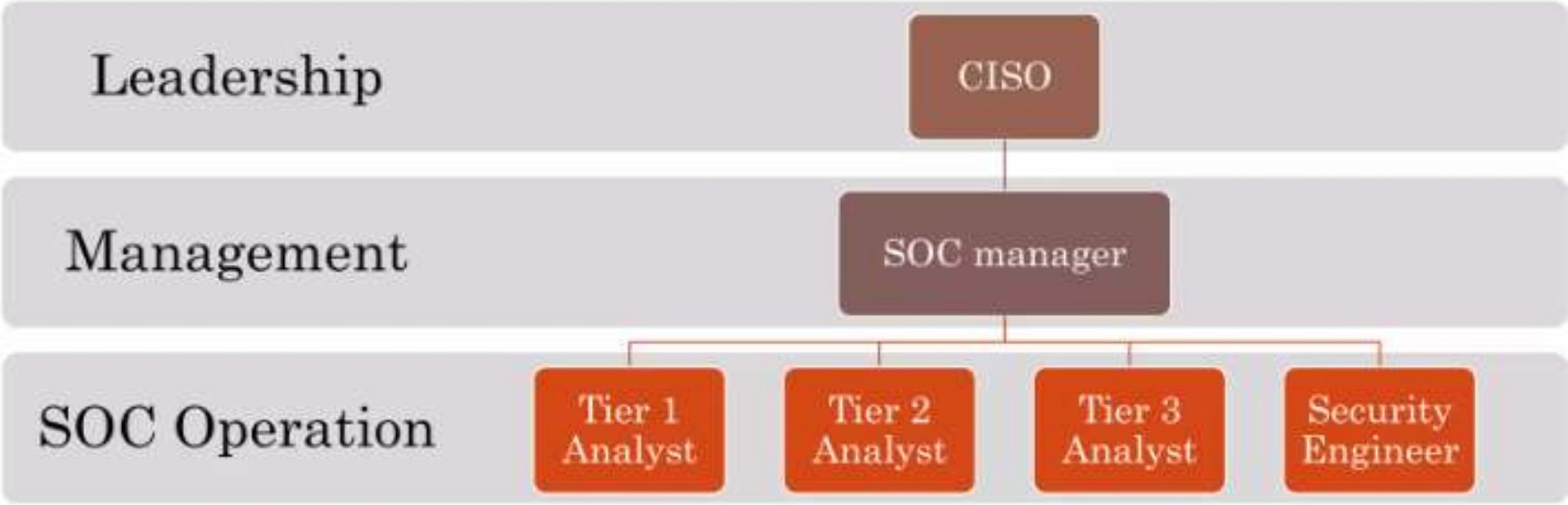
- Detect
  - Install and update anti-virus, anti-spyware and other anti-malware programs.
  - Maintain and monitoring Logs.
- Respond
  - Develop a plan for disasters for information security incidents.
- Recovery
  - Make full pack up of important data and information.

SECURITY OPERATION CENTER (SOC)

monitor, prevent, detect, investigate, and respond to cyber threats around the clock

SECURITY OPERATION CENTER (SOC) *(CONT.)*

- **SOC Team**

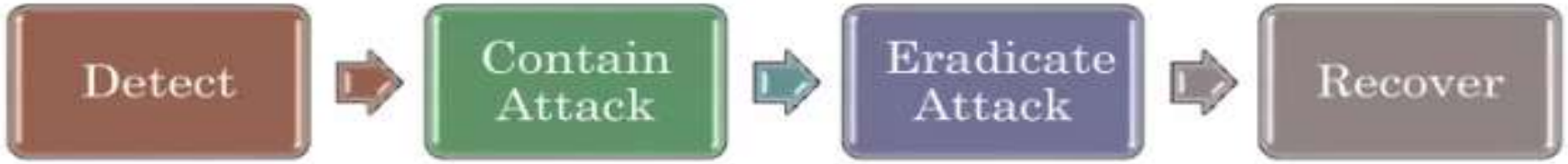


SECURITY OPERATION CENTER (SOC) *(CONT.)*

- **Tier 1 Analyst (Alert Investigator) :**
  - Monitor SIEM alerts.
  - Manages and configures security Monitoring Tools.
  - Alert priority.
  - Perform triage to confirm real security incident is taking place.
- **Tier 2 Analyst (Incident responder):**
  - Receives Incident and performs deep analysis.
  - Correlate with threat intelligence to identify threat actor.
  - Nature of the attack.
  - Data and systems affected.
  - Decide strategy for containment.
  - Remediation and recovery.

SECURITY OPERATION CENTER (SOC) *(CONT.)*

- **Tier 3 Analyst (SME / Threat Hunters):**
  - Vulnerability assessment.
  - Penetration testing.
  - Threat intelligence.
  - Threat Hunters who hunts threat which found their way into the network.
  - Unknown vulnerabilities and security gaps.
  - When major incident occurs join with Tier 2 analyst in responding and containing it.



SECURITY OPERATION CENTER (SOC) *(CONT.)*

- **Security Engineers (Platform Management):**
  - Automated Tools.
  - Integration between security controls and SIEM.

- **SOC manager:**
  - Responsible for hiring and training SOC staff.
  - Manage resources. (Metrics)
  - Manage team when responding to critical security incident.

SECURITY OPERATION CENTER (SOC) *(CONT.)*

- **SOC process**
  - Log source management
  - SIEM management
  - Use case management
  - Playbook management
  - Event management
  - Incident management
  - Vulnerability management

SOC PLATFORM (TOOLS)

- **SIEM** : Security Information and Event Management
- **SOAR** : Security Orchestration, Automation and Response
- **VMDR** : Vulnerability Management, Detection and response
- **NDR** : Network Detection and Response
- **EDR** : End-point Detection and response
- **TIP** : Threat Intelligence Platform
- **OST** : Offensive Security Tools

SKILLS NEEDED IN SOC

- **Tier 1 Analyst**
  - 2-3 years of professional experience.
  - Very good routing & switching knowledge.
  - Good system administration knowledge.
  - Understanding security system functions.
  - Knowledge of SIEM event management.
  - Certificates: CompTIA Cyber Security Analyst (CSA), SANS GMON

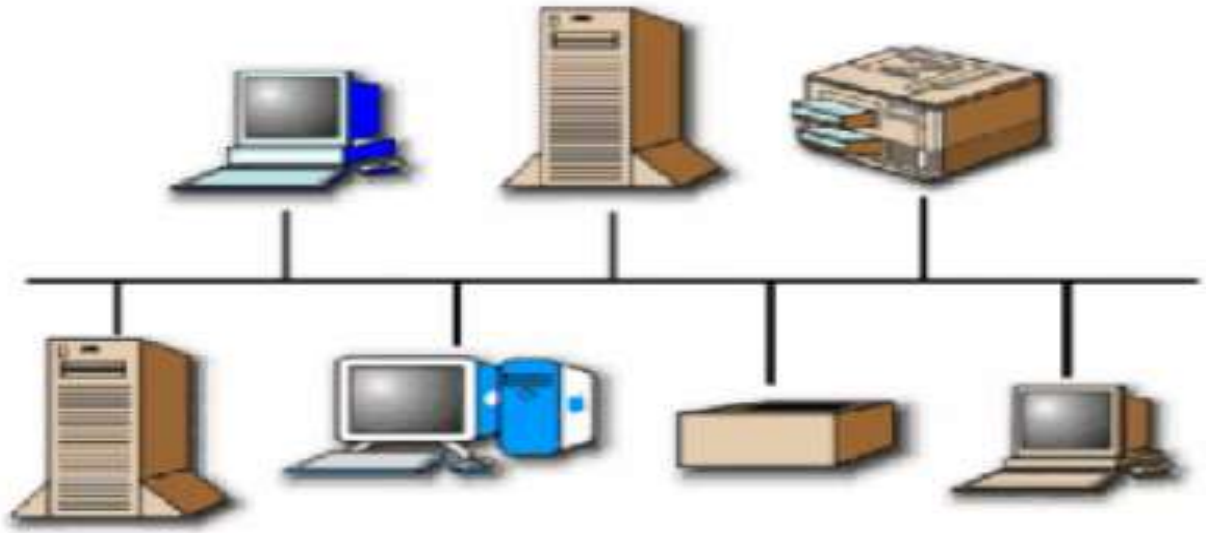
TYPES OF SOC

Dedicated SOC	Classic SOC with dedicated full time staff, operated fully in house 24/7/365 operations.
Distributed SOC	Some full time staff and some part time, typically operates 8x5 in each region
Multifunctional SOC / NOC	Dedicated team which perform both functions of a network operation center and a SOC
Fusion SOC	Traditional SOC combined with new functions such as threat intelligence, operational technology
Command SOC / Global SOC	Coordinates other SOCs in global enterprise provide threat intelligence, situational awareness and guidance
Virtual SOC	No dedicated facility, part time members usually reactive and activated by security incident
Managed SOC	Many organizations turned to MSSP Managed Security Service Providers to provide SOC services on outsourced basis

BASICS OF NETWORKING

What is a Computer Network?

A network is a collection of computers, printers, routers, switches, and devices that are able to communicate with each other over some transmission media.



Types of Networks

There are two basic types of networks currently in existence:

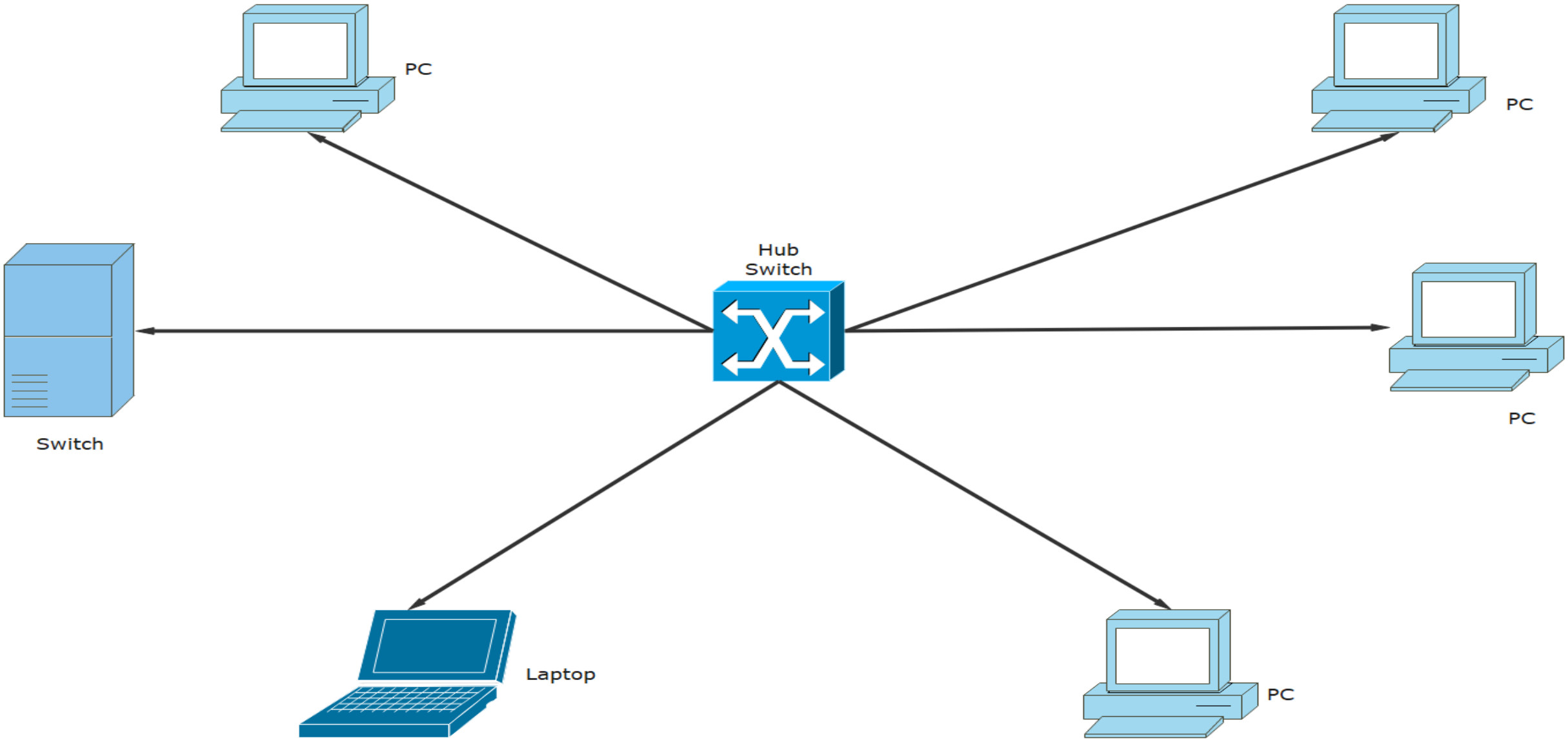
A *Local Area Network* (LAN)

A *Wide Area Network* (WAN)

TYPES OF AREA NETWORKS - LAN, MAN AND WAN

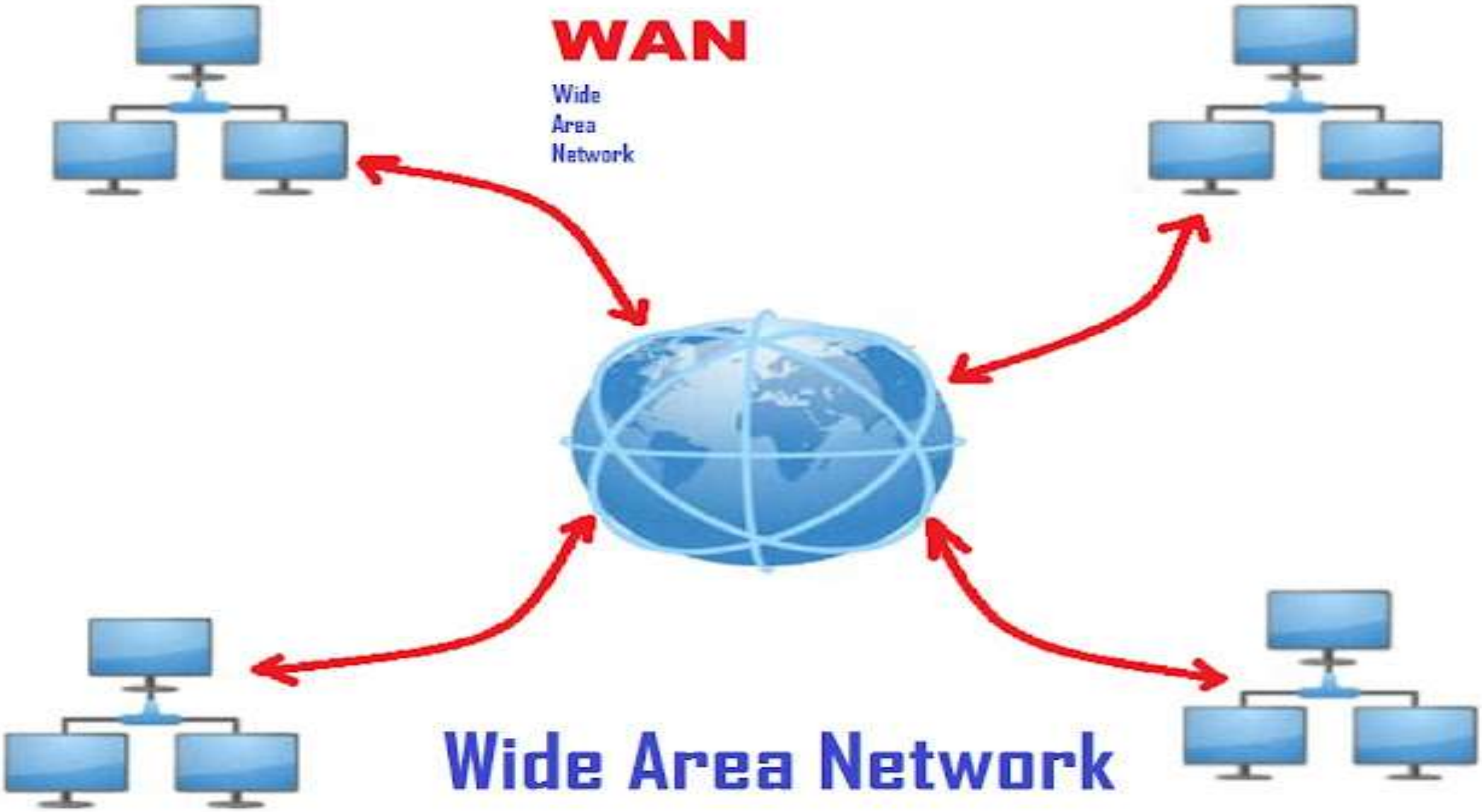
- **Local Area Network (LAN) –**
- LAN or Local Area Network connects network devices in such a way that personal computers and workstations can share data, tools, and programs. The group of computers and devices are connected together by a switch, or stack of switches, using a private addressing scheme as defined by the TCP/IP protocol. Private addresses are unique in relation to other computers on the local network. Routers are found at the boundary of a LAN, connecting them to the larger WAN.

LAN Diagram



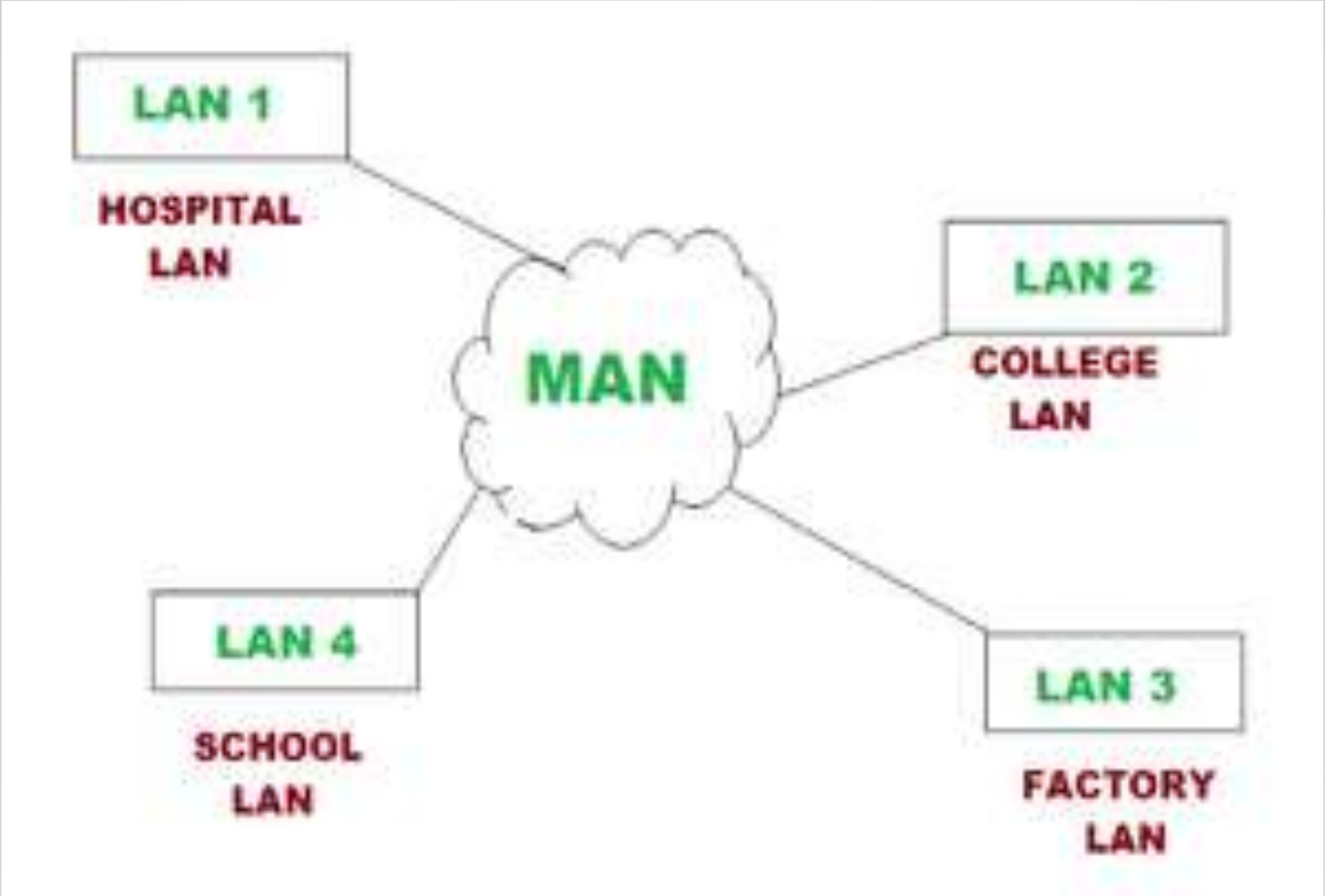
WIDE AREA NETWORK (WAN) –

- WAN or Wide Area Network is a computer network that extends over a large geographical area, although it might be confined within the bounds of a state or country. WAN has a range of above 50 km. A WAN could be a connection of LAN connecting to other LANs via telephone lines and radio waves and may be limited to an enterprise (a corporation or an organization) or accessible to the public. The technology is high-speed and relatively expensive.


WAN
Wide
Area
Network
Wide Area Network

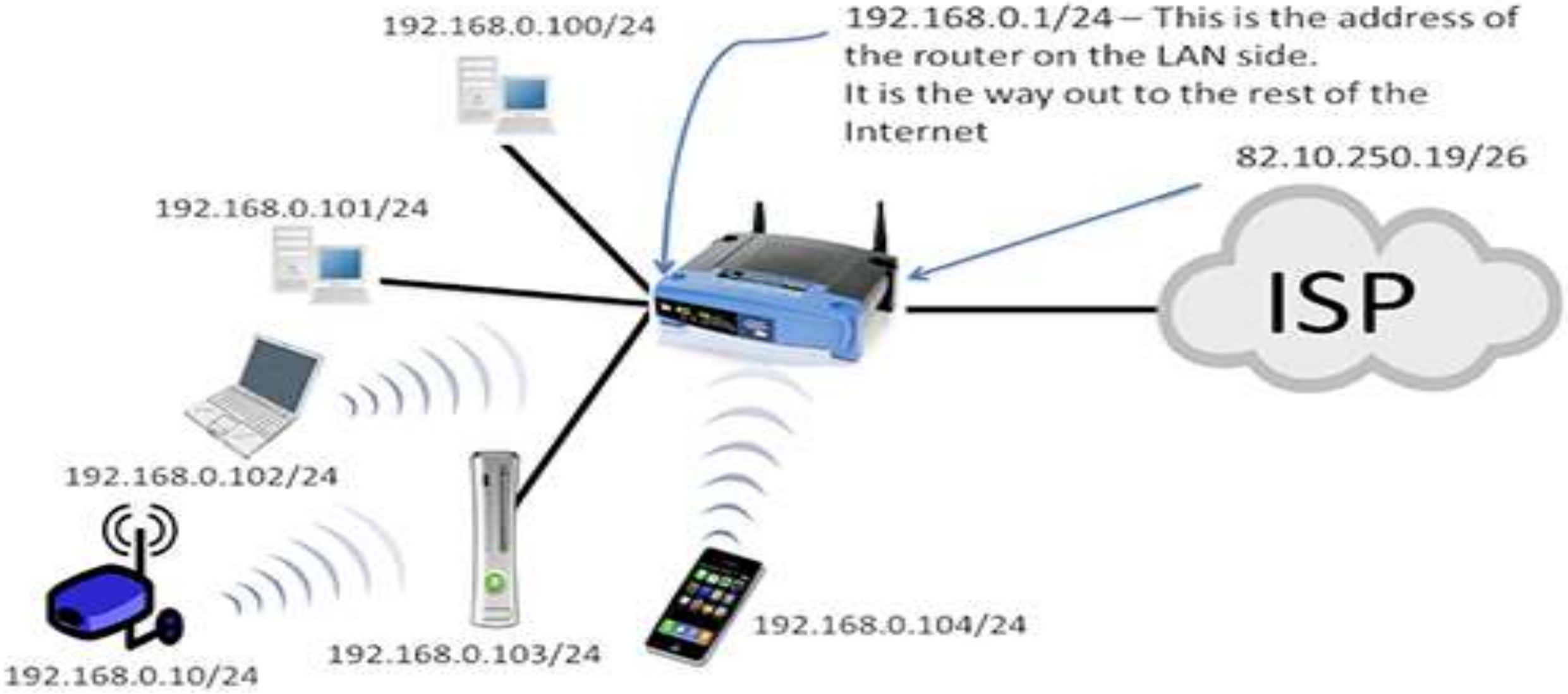
METROPOLITAN AREA NETWORK (MAN) –

- MAN or Metropolitan area Network covers a larger area than that covered by a LAN and a smaller area as compared to WAN. MAN has a range of 5-50km. It connects two or more computers that are apart but reside in the same or different cities. It covers a large geographical area and may serve as an ISP (Internet Service Provider). MAN is designed for customers who need high-speed connectivity. Speeds of MAN range in terms of Mbps. It's hard to design and maintain a Metropolitan Area Network.


LAN 1
HOSPITAL LAN
LAN 2
COLLEGE LAN
MAN
LAN 4
SCHOOL LAN
LAN 3
FACTORY LAN

IP (INTERNET PROTOCOL)

IP (INTERNET PROTOCOL) IS THE FUNDAMENTAL PROTOCOL USED FOR ADDRESSING AND ROUTING DATA PACKETS ACROSS NETWORKS, INCLUDING THE INTERNET. IT ENSURES DATA REACHES THE CORRECT DESTINATION USING UNIQUE NUMERICAL ADDRESSES ASSIGNED TO DEVICES.



IPv4 (Internet Protocol version 4)

IPv4 is the fourth version of IP and the most widely used protocol for internet communication

Address Length: 32-bit (e.g., 192.168.1.1)
Address Space: Approximately 4.3 billion addresses (2^{32})

-
- ip = 255:255:255:255 (8bit per octet)

IPv6 (Internet Protocol version 6)

IPv6 is the successor to IPv4, designed to overcome address exhaustion and provide improved performance and security.

Address Length: 128-bit (e.g., 2001:db8::ff00:42:8329)
Address Space: 3.4×10^{38} addresses (virtually unlimited)
Notation: Hexadecimal format with colons (e.g., 2001:0db8:85a3::8a2e:0370:7334)

Subnetting: Breaking Down Networks Efficiently

Subnetting is the process of dividing a larger IP network into smaller

Prevents wasting IP addresses in large networks.

Segments networks, restricting access between different subnets.

Subnetting in IPv4

IPv4 addresses consist of **two parts**:

Network Portion - Identifies the network.

Host Portion - Identifies individual devices (hosts) in the network.

Subnet Mask

A **subnet mask** determines which portion of an IP address represents the network.

Consider an IP **192.168.1.0/26**

192.168.00000001.0/26

Subnet mask

First 26 bit reserved for a network

Classful vs. Classless Addressing in IP Networks

1. Classful Addressing (Traditional Approach)

Classful addressing was the original IP addressing method

IP Address Classes in Classful Addressing:

A company needing 300 IPs had to use a Class B address (65,534 addresses), wasting thousands of IPs.

Class	First Octet Range	Default Subnet Mask	Number of Hosts (Usable)	Usage
A	1 - 126	255.0.0.0 (/8)	~16.7 million	Large networks
B	128 - 191	255.255.0.0 (/16)	~65,000	Medium networks
C	192 - 223	255.255.255.0 (/24)	254	Small networks
D	224 - 239	N/A	N/A	Multicast
E	240 - 255	N/A	N/A	Experimental

2. Classless Addressing (CIDR – Modern Approach)

Classless Inter-Domain Routing (CIDR) was introduced to overcome the limitations of classful addressing.

Example of CIDR and VLSM:

Instead of allocating a full **Class B (65,534 addresses)** to a company needing **1000 IPs**, CIDR allows:

- Assigning 192.168.1.0/22 → 1022 usable addresses.
- Another department with **500 IPs** gets 192.168.2.0/23 → 510 usable addresses.

CIDR Notation Example:

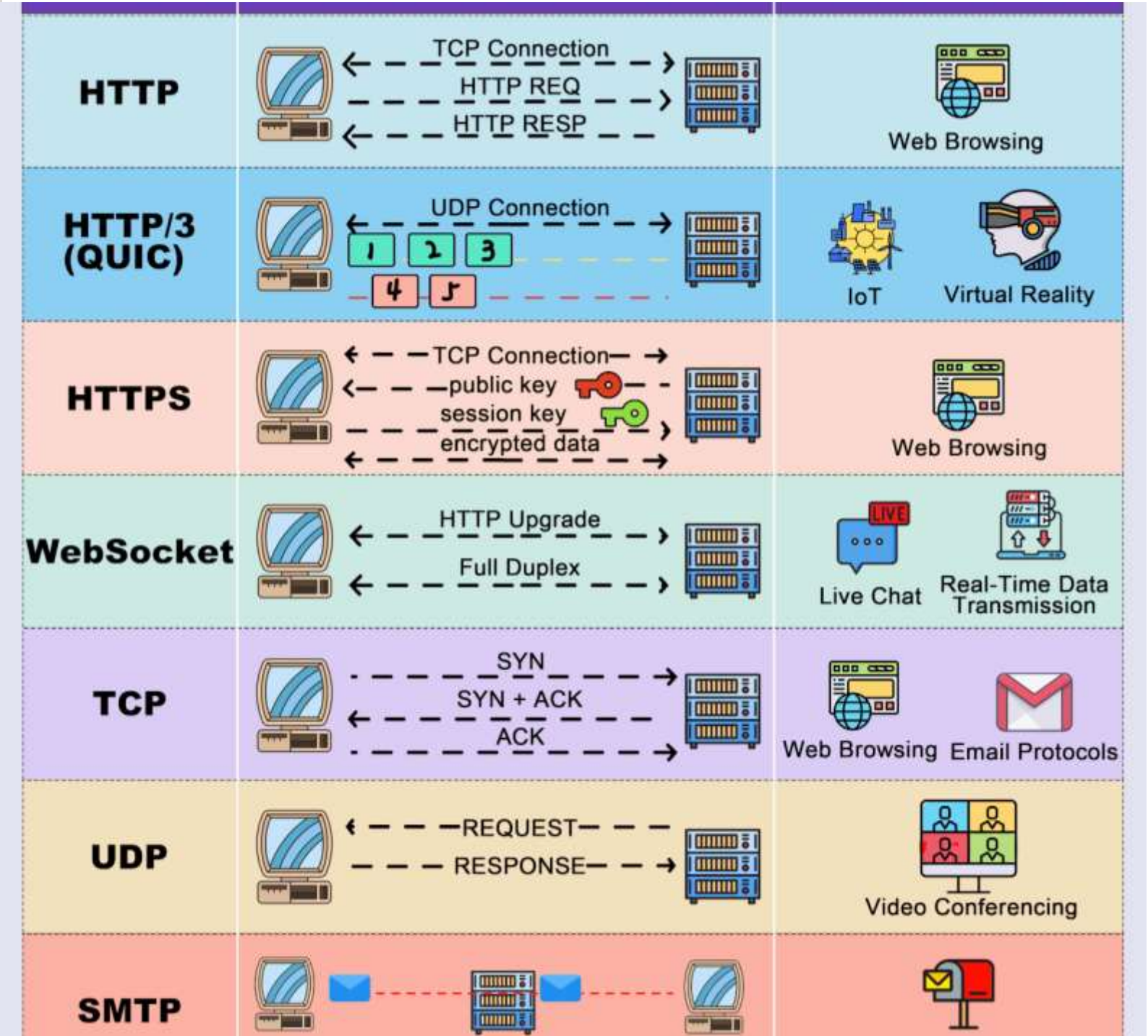
- 192.168.10.0/24 → 256 total addresses, 254 usable.
- 192.168.10.0/26 → 64 total addresses, 62 usable.

.Well-Known Ports (0-1023): Reserved for widely used services and protocols. Examples include:

- HTTP (Port 80)
- HTTPS (Port 443)
- FTP (Port 21)

2. Registered Ports (1024-49151): Assigned to specific applications by organizations. These ports are not as universally recognized but are commonly used.

3. Dynamic/Private Ports (49152-65535): Typically used for temporary connections and can be utilized by any process.



INTRODUCTION TO NETWORK PROTOCOLS

DEFINITION: NETWORK PROTOCOLS ARE STANDARDIZED RULES FOR DATA COMMUNICATION ACROSS NETWORKS.
IMPORTANCE: ENSURES RELIABLE, SECURE, AND EFFICIENT COMMUNICATION BETWEEN DEVICES.
CATEGORIES:

COMMUNICATION PROTOCOLS (TCP, UDP, HTTP, FTP)
SECURITY PROTOCOLS (SSL/TLS, IPSEC)
ROUTING PROTOCOLS (OSPF, BGP)

TCP header 20-60 byte

<table>
<tr><td colspan="8">Source Port(16)</td><td>Destination Port(16)</td></tr>
<tr><td colspan="9">Sequence Number(32 bits)</td></tr>
<tr><td colspan="9">Acknowledgement Number(32 bits)</td></tr>
<tr><td>Header Length (4 bits)</td><td>6 Bits Reserved</td><td>URG (1 bit)</td><td>ACK (1 bit)</td><td>RST (1 bit)</td><td>PSH (1 bit)</td><td>SYN (1 bit)</td><td>FIN (1 bit)</td><td>Window Size(16 bits)</td></tr>
<tr><td colspan="8">Checksum(16 bits)</td><td>Urgent Pointer(16 bits)</td></tr>
<tr><td colspan="9">Options/Optional data(0-40 bytes)</td></tr>
</table>

HTTP & HTTPS:WEB BROWSING PROTOCOLS

HTTP (HYPERTEXT TRANSFER PROTOCOL)80

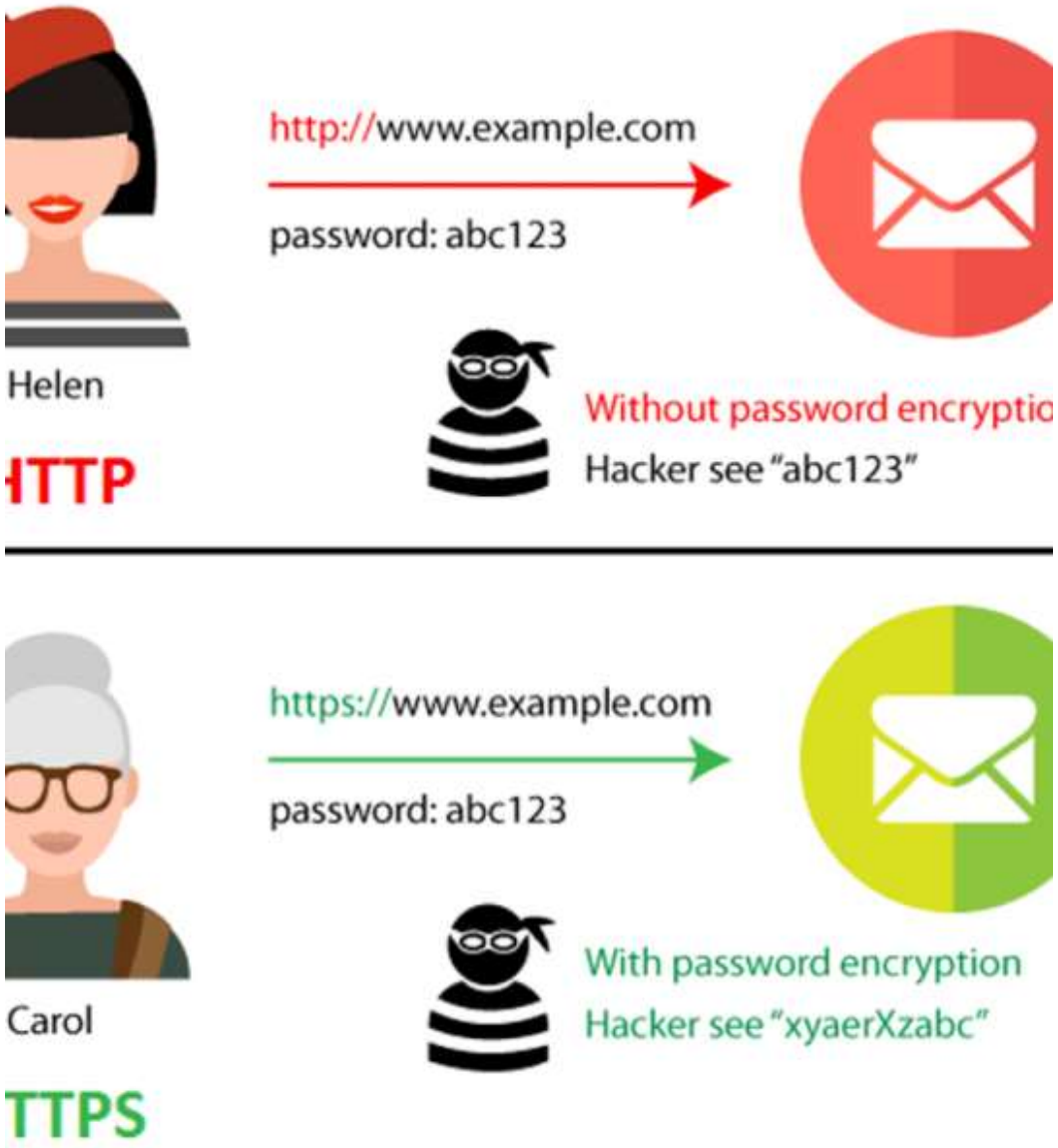
HTTP IS A **STATELESS, APPLICATION-LAYER PROTOCOL** USED FOR TRANSMITTING HYPERTEXT, IMAGES, VIDEOS, AND OTHER WEB CONTENT.

HTTPS (HYPERTEXT TRANSFER PROTOCOL SECURE)443

HTTPS IS THE SECURE VERSION OF HTTP, USING **SSL/TLS ENCRYPTION** TO PROTECT DATA.

HOW HTTPS WORKS:

1. **TLS HANDSHAKE:** BROWSER AND SERVER ESTABLISH A SECURE CONNECTION USING **SSL/TLS CERTIFICATES**.
2. **ENCRYPTED COMMUNICATION:** ALL DATA BETWEEN THE CLIENT AND SERVER IS **ENCRYPTED**, PREVENTING INTERCEPTION.



IPV4 (INTERNET PROTOCOL VERSION 4)

- **Format**: 32-bit, written in **decimal**, e.g., 192.168.1.1
- **Range**: 0.0.0.0 to 255.255.255.255
- **Total Addresses**: ~4.3 billion
- **Classes**:
  - **Class A**: 1.0.0.0 to 126.255.255.255 (Large networks)
  - **Class B**: 128.0.0.0 to 191.255.255.255 (Medium networks)
  - **Class C**: 192.0.0.0 to 223.255.255.255 (Small networks)
- **Private Ranges**:
  - 10.0.0.0/8
  - 172.16.0.0/12
  - 192.168.0.0/16
- **Example**: 192.168.1.10 (private IP used in homes/offices)

IPV6 (INTERNET PROTOCOL VERSION 6)

- **Format**: 128-bit, written in **hexadecimal**, e.g., 2001:0db8:85a3:0000:0000:8a2e:0370:7334
- **Total Addresses**: 340 undecillion+ (almost infinite)
- Designed to **replace IPv4** due to exhaustion.
- Supports auto-configuration and better **security (IPSec)**.
- **Zero compression**: :: used to shorten consecutive 0s.
- **Example**: fe80::1 (link-local address)

PORTS & PROTOCOLS

- A **port** is a **virtual point** where network connections start and end.
- Used to identify **specific services** running on a device.
- Ranges:
  - **0–1023**: Well-known ports (standard services)
  - **1024–49151**: Registered ports (apps/services)
  - **49152–65535**: Dynamic/private (ephemeral)

COMMON PORT PROTOCOLS

20	FTP	File Transfer	TCP
21	FTP	Control Channel	TCP
22	SSH	Secure Shell	TCP
23	Telnet	Remote Login	TCP
25	SMTP	Email Sending	TCP
53	DNS	Domain Resolution	UDP (mainly), TCP (for zone transfer)
67/68	DHCP	IP Assignment	UDP
80	HTTP	Web Browsing	TCP
110	POP3	Email Receiving	TCP
123	NTP	Time Sync	UDP
143	IMAP	Email Receiving	TCP
161	SNMP	Network Mgmt	UDP
443	HTTPS	Secure Web	TCP
445	SMB	File Sharing	TCP
3389	RDP	Remote Desktop	TCP

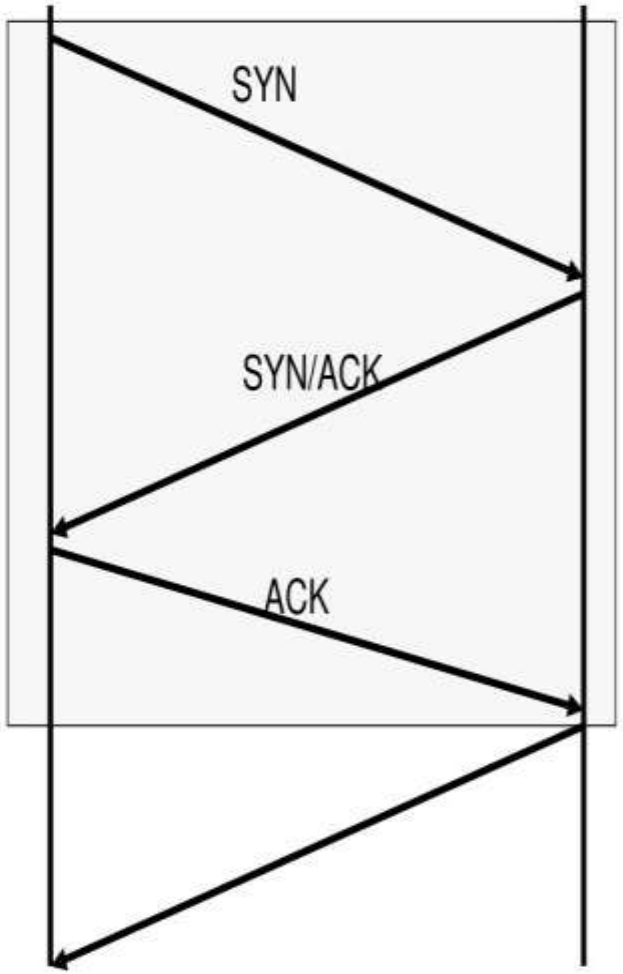
WHAT IS TCP (TRANSMISSION CONTROL PROTOCOL)?

- **Connection-oriented** protocol (reliable).
- Establishes a **3-way handshake** before communication.
- Ensures:
  - **Packet delivery**
  - **Data ordering**
  - **Error checking**
- Used where reliability is critical: **web, file transfer, email, remote access**.

TCP 3-WAY HANDSHAKE:

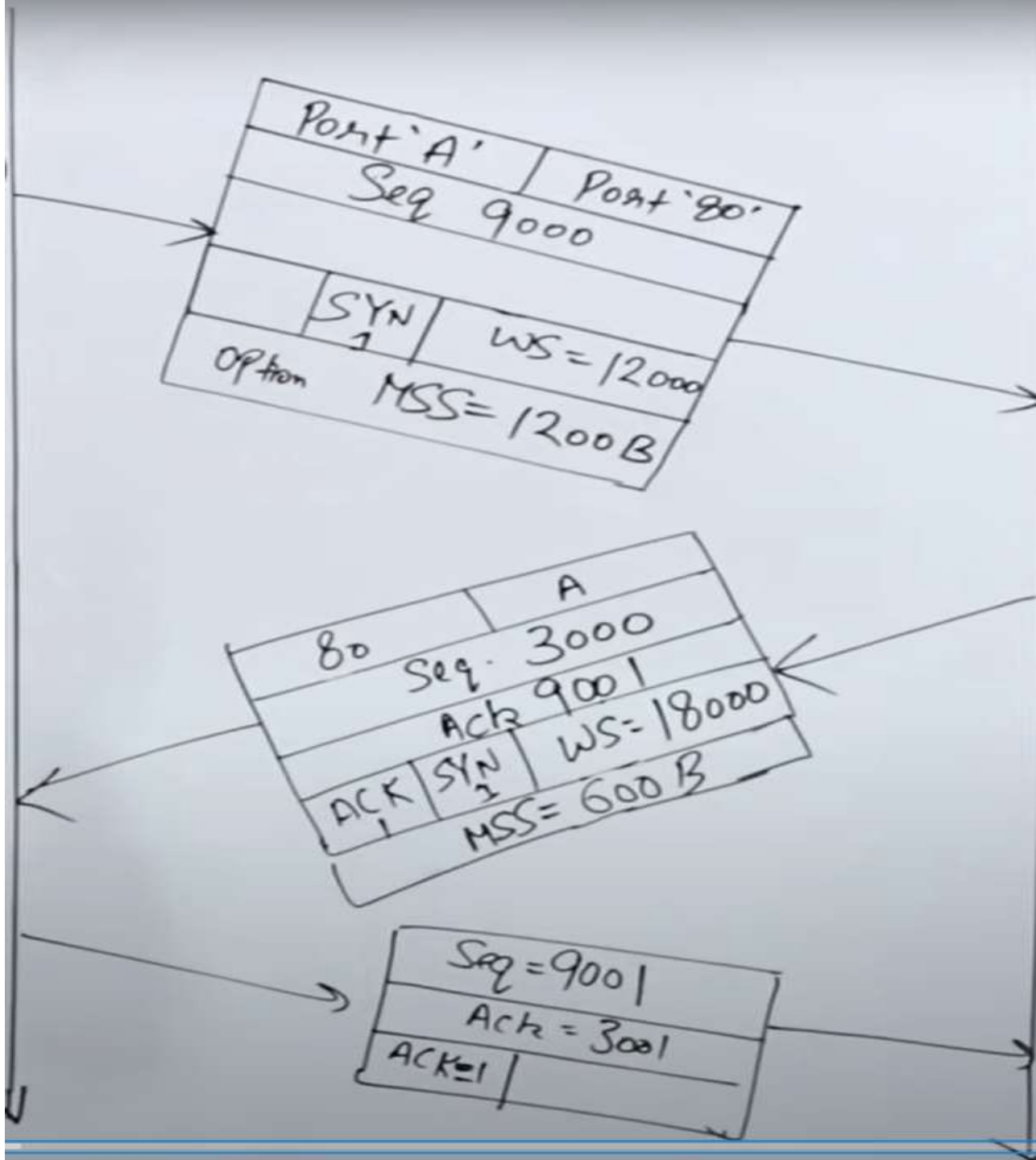
- **SYN** → client requests connection
- **SYN-ACK** → server acknowledges
- **ACK** → client confirms

TCP Handshake



Connection Establishment (Handshake)

Transmission of data



WHAT IS UDP (USER DATAGRAM PROTOCOL)?

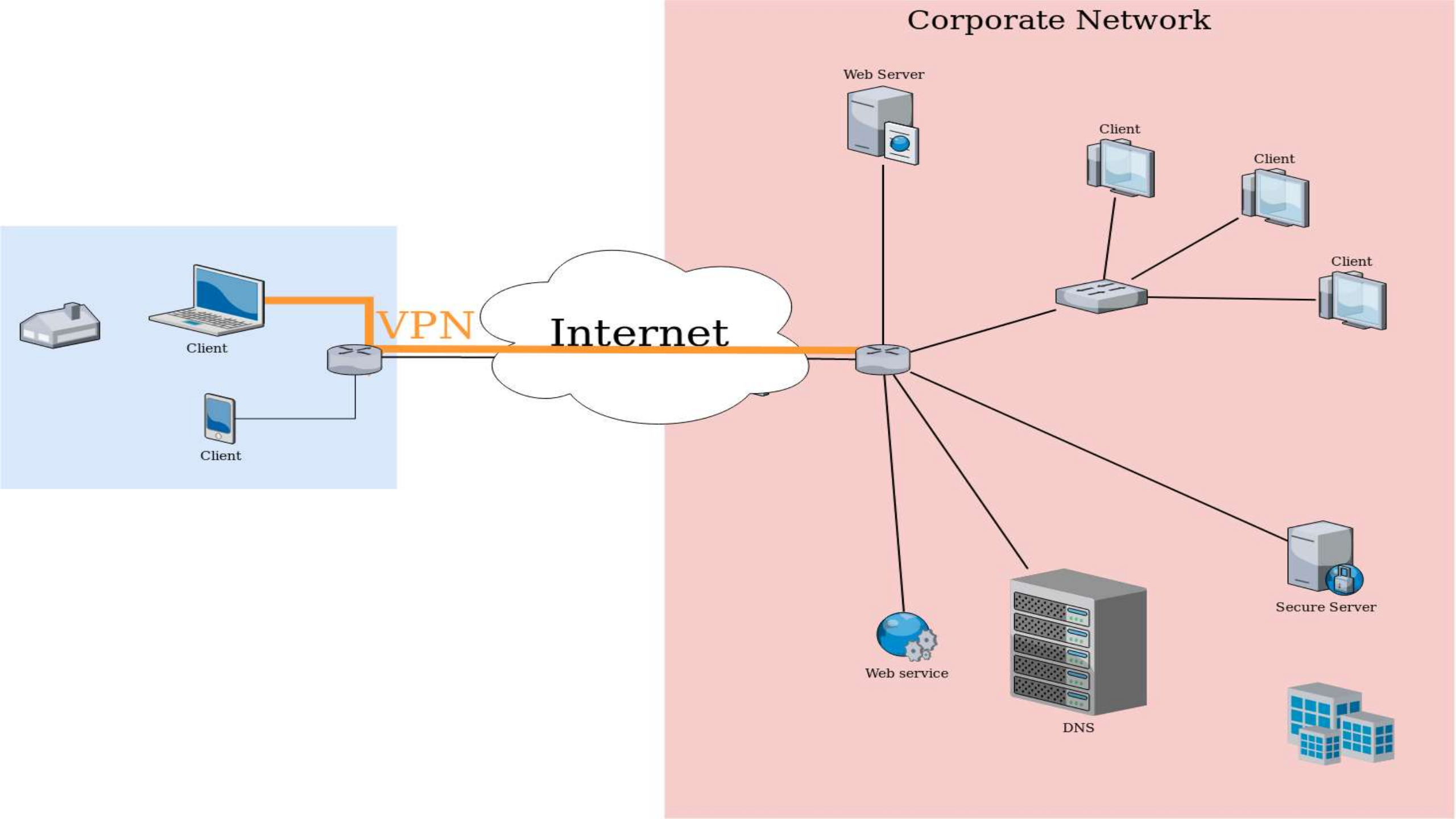
- **Connectionless** protocol (unreliable but fast).
- No handshake or acknowledgment.
- Less overhead, faster than TCP.
- Used where speed matters: **streaming, gaming, DNS, VoIP**

SOC ANALYST FOCUS AREA ON TCP & UDP

Security Monitoring	Track login attempts, file transfer	Monitor flooding attacks, DNS abuse
Detection	More stateful, easier to trace	Harder to trace, often spoofed
Attacks	Brute-force, SYN floods	DNS amplification, UDP floods
Mitigation	Firewall rules by state/port	Rate limiting, deep packet inspection

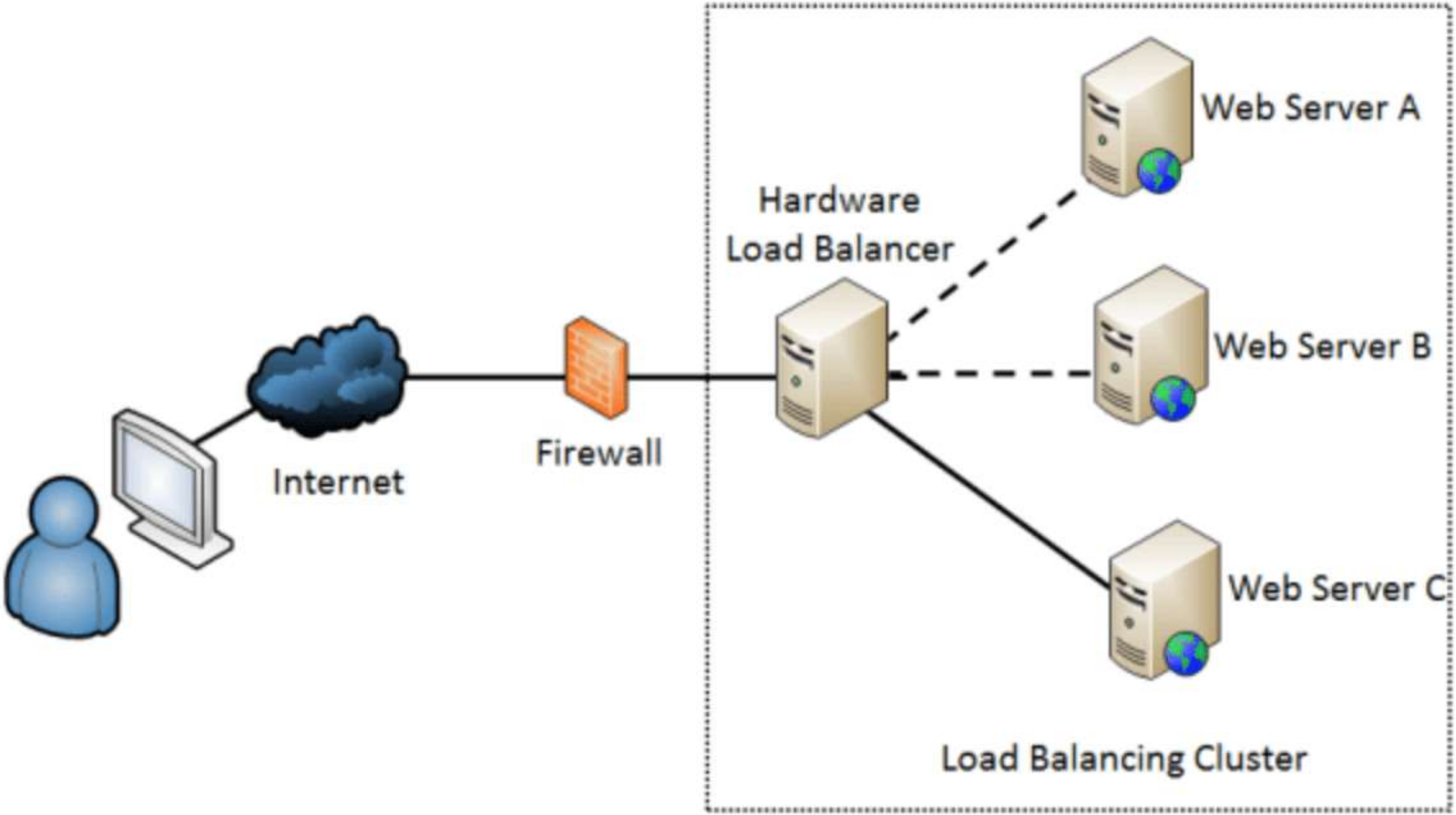
VPN (VIRTUAL PRIVATE NETWORK)

- Creates encrypted connection over public network.
- Used for secure remote access.
- SOC Focus: VPN logs, user access monitoring, anomaly detection.


Corporate Network
Web Server
Client
Client
Client
Client
VPN
Internet
Client
Web service
DNS
Secure Server

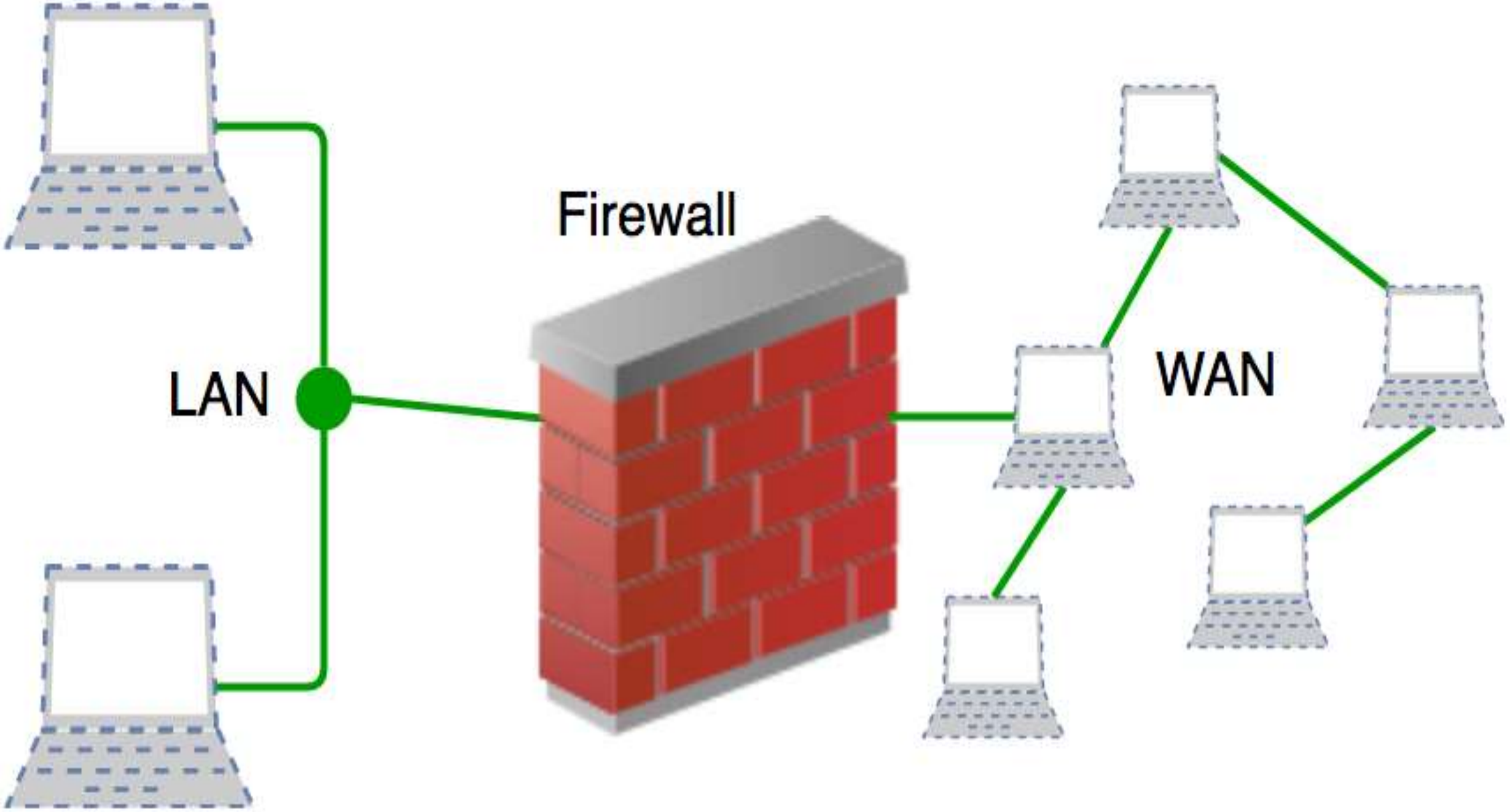
LOAD BALANCER

- Distributes network traffic across multiple servers.
- Prevents server overload and increases availability.
- Can be hardware or software-based.

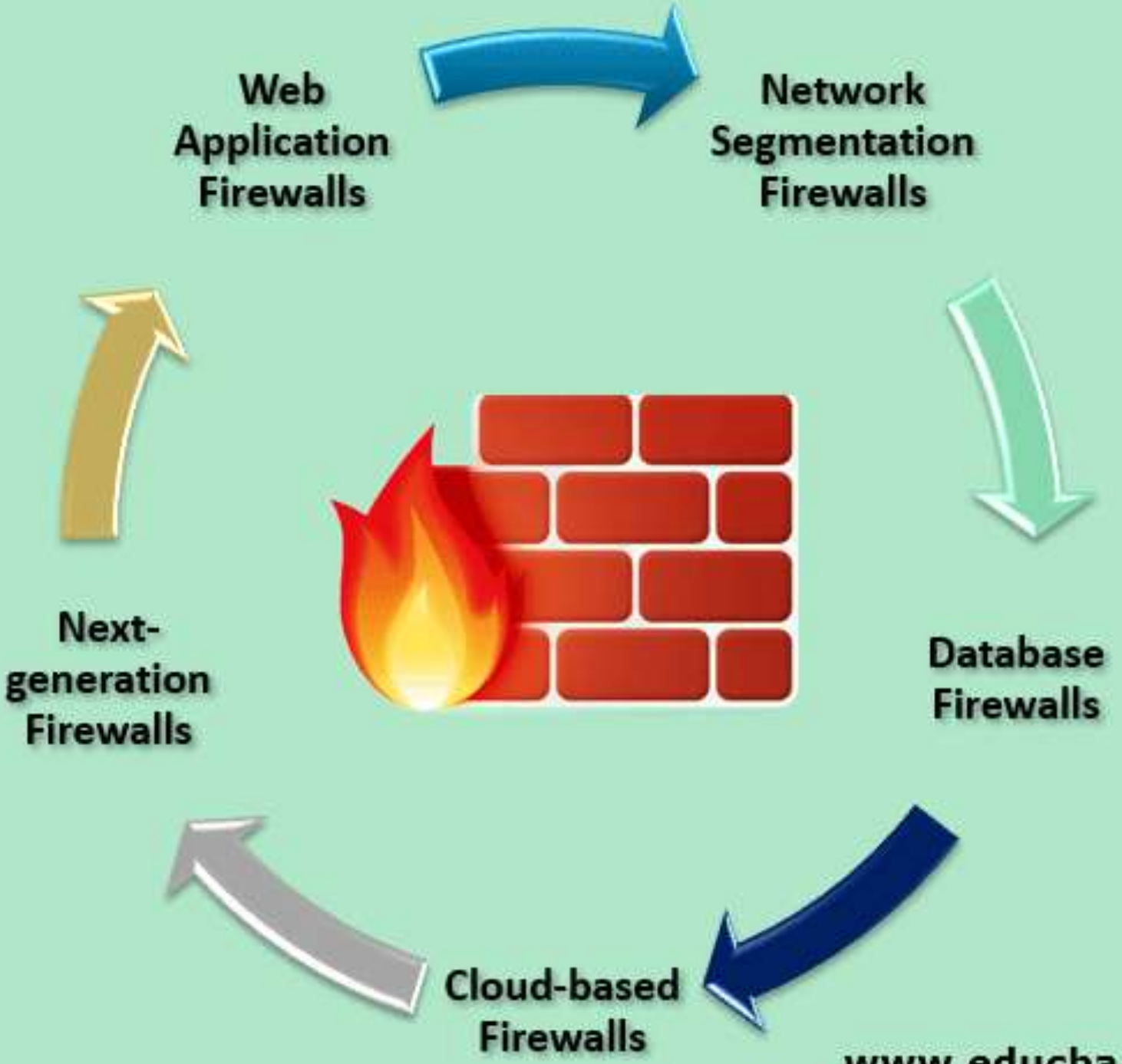

Web Server A
Hardware
Load Balancer
Web Server B
Internet
Firewall
Web Server C
Load Balancing Cluster

FIREWALL

- A firewall is a network security device either hardware or software-based which monitors all incoming and outgoing traffic and based on a defined set of security rules it accepts, rejects, or drops that specific traffic. It acts like a security guard that helps keep your digital world safe from unwanted visitors and potential threats.
- **Accept:** allow the traffic
- **Reject:** block the traffic but reply with an "unreachable error"
- **Drop:** block the traffic with no reply

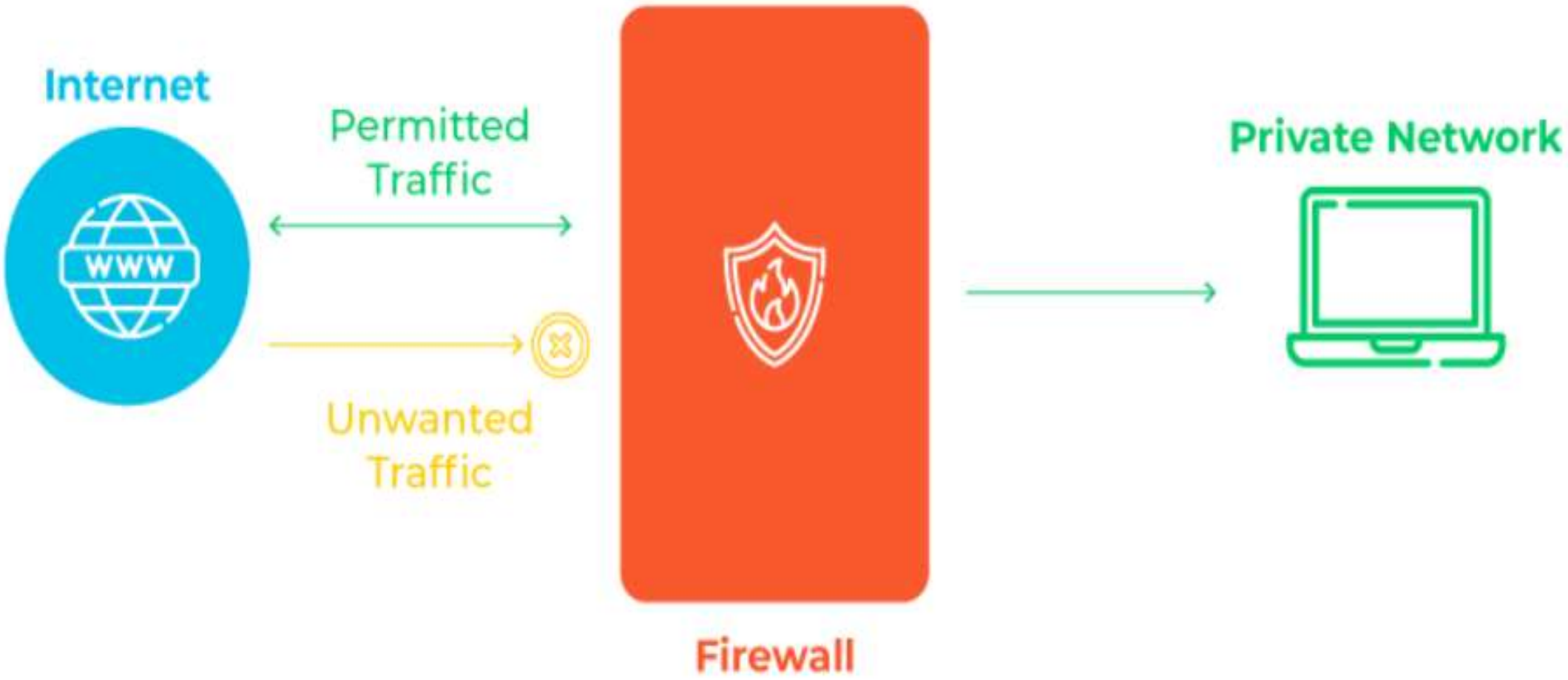

Firewall
LAN
WAN

Types of Firewalls



www.educba.com

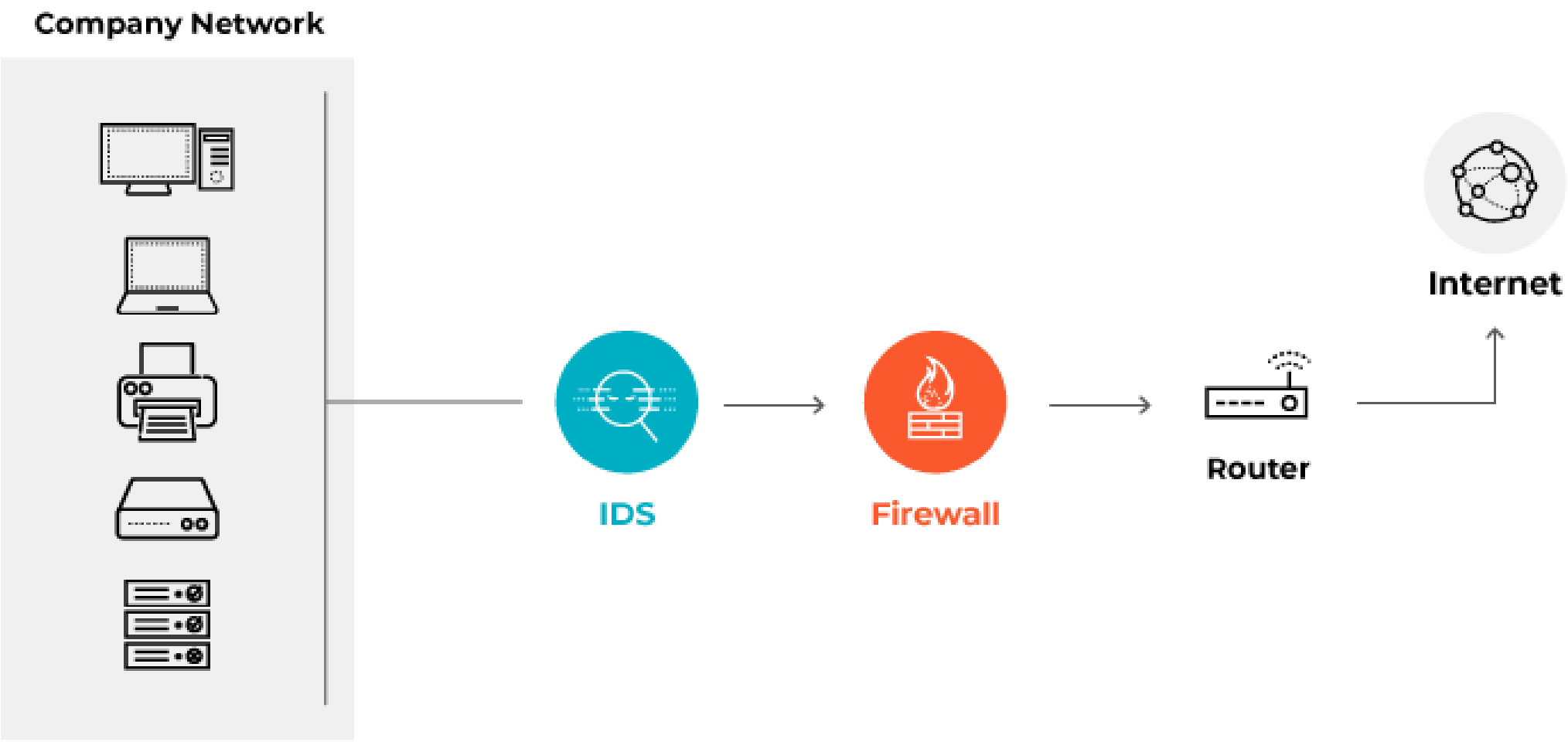
How Firewalls Work



WHAT IS AN INTRUSION DETECTION SYSTEM (IDS)?

- An intrusion detection system (IDS) identifies potential threats and weaknesses in networked systems. An IDS examines network traffic, alerting administrators to suspicious activities without intervening in data transmission.
- IDS is either a hardware or software program that analyzes incoming network traffic for malicious activities or policy breaches (network behavior analysis) and issues alerts when they are detected. It detects real-time traffic and searches for attack signatures or traffic patterns, then sends out alarms. Unlike IPS, a network Intrusion Detection System is not in line with the data path, so it can only alert and alarm on detection of anomalies.

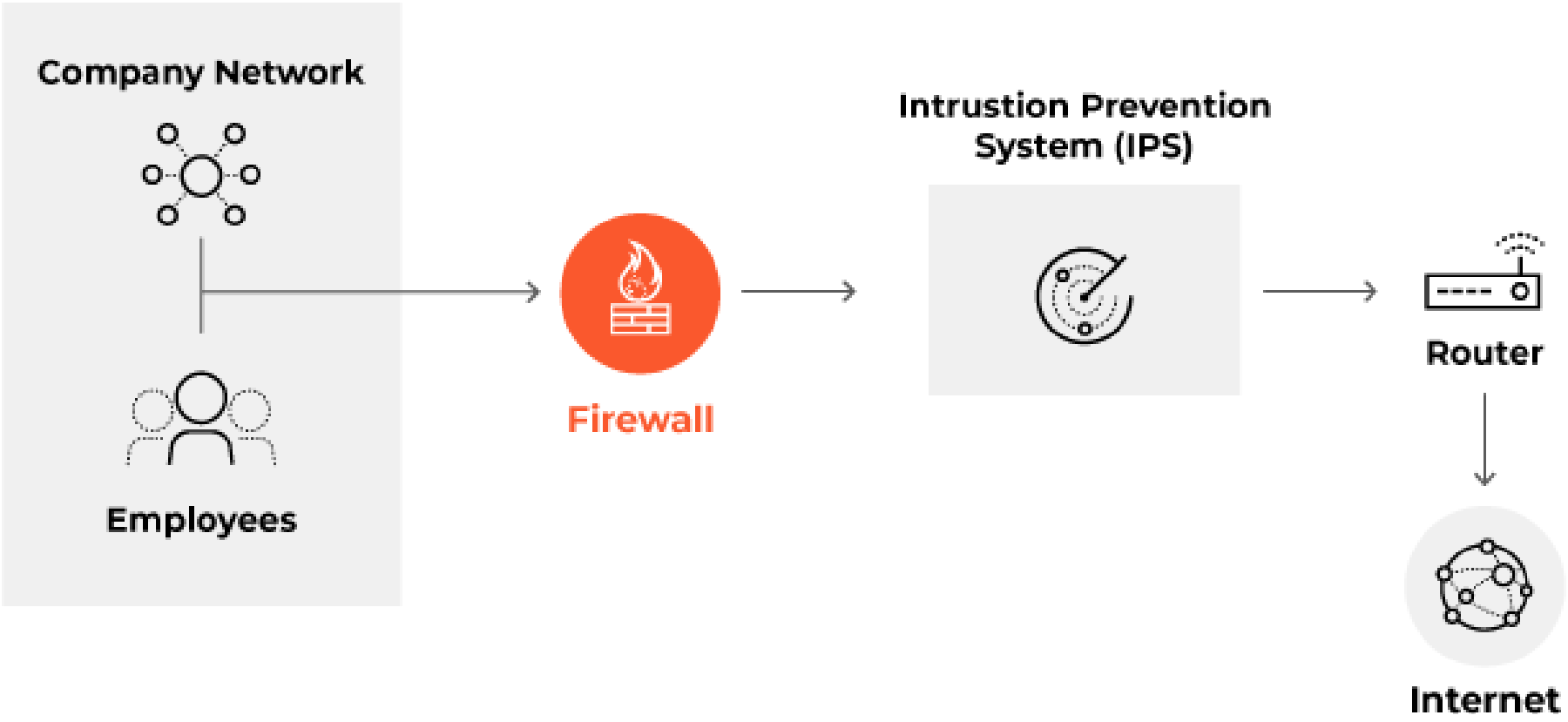
Intrusion Detection System



INTRUSION PREVENTION SYSTEM (IPS)

- IPS is a device that inspects, detects, classifies, and proactively prevents harmful traffic. It examines real-time communications for attack patterns or signatures and then blocks attacks when they have been detected. Placement and configuration in inline mode and generally being in Layer 2 after the firewall. In inline mode, traffic passes into one of the device's ethernet ports and out of the other.
- Intrusion Prevention System must work efficiently to avoid decreasing network performance. It must be quick because exploits might occur anytime. To eliminate threats and false positives, the IPS must detect and respond accurately.
- Some of the actions of IPS include:
- Alerting network administrators (anomaly-based detection)
- Dropping the malicious traffic
- Denying traffic from the source address
- Reset the connection

Intrusion Prevention System



Intrusion Prevention System vs. Intrusion Detection system vs. Firewall

Parameter	Firewall	IPS	IDS
Purpose	A network security device that filters incoming and outgoing traffic based on predetermined security rules.	A device that inspects and prevents identified threats in real-time by analyzing traffic.	A system that monitors network or system activities for malicious actions or policy violations.
Operation	Filters traffic based on rules applied to addresses and port numbers.	Examines traffic for real-time attacks and intervenes to stop them on detection.	Observes traffic and looks for attack patterns or anomalies, generating alerts.
Configuration mode	Operates in either inline or transparent mode at the network's boundary.	Typically inline, positioned after the firewall within the network layer.	Generally, operates in monitoring mode, not inline with traffic flow.
Traffic path	Should be the primary route of network traffic.	Placed after the firewall to examine filtered traffic.	Analyzes traffic after it has passed through the firewall.
Placement	Located at the network perimeter as the initial defense line.	Positioned right after the firewall, before the internal network.	Resides within the network, typically after the IPS for deeper traffic analysis.
Response to unauthorized traffic	Blocks or permits traffic based on rule evaluation.	Actively prevents the progression of detected threats.	Issues alerts or alarms upon detecting suspicious activities.